

THE DEFORMED HERMITIAN-YANG-MILLS EQUATION: SOLVABILITY AND STABILITY

1. THE EQUATION AND SOLVABILITY

Let X be a compact connected Kähler manifold of dimension $n \geq 2$, $A \in H^{1,1}(X, \mathbb{R})$, and $B \in \mathcal{K}_X$, the Kähler cone. For a Kähler form $\omega \in B$, the dHYM equation seeks a smooth closed real $(1, 1)$ form $\alpha \in A$ such that

$$\Theta_\omega(\alpha) = \sum_{j=1}^n \arctan \lambda_j(\omega^{-1}\alpha) = \hat{\theta}, \quad \arctan \lambda_j \in (-\pi/2, \pi/2).$$

The phase $\hat{\theta} \in \mathbb{R}$ is *supercritical* when $(n-2)\pi/2 < \hat{\theta} < n\pi/2$, and *compatible* when $\int_X (B + \sqrt{-1}A)^n \in e^{\sqrt{-1}\hat{\theta}} \mathbb{R}_{>0}$. Compatibility determines it only modulo 2π ; the equation uses its real lift.

Theorem 1.1 (Jacob–Yau [6]). *Let (X, ω) be a compact connected Kähler surface, $B = [\omega]$, and $A \in H^{1,1}(X, \mathbb{R})$. Suppose*

$$0 < \hat{\theta} < \pi, \quad \int_X (B + \sqrt{-1}A)^2 \in e^{\sqrt{-1}\hat{\theta}} \mathbb{R}_{>0}.$$

There is a smooth closed $\alpha \in A$ with $\Theta_\omega(\alpha) = \hat{\theta}$ if and only if

$$A + \cot \hat{\theta} B \in \mathcal{K}_X.$$

Theorem 1.2. *Let (X, ω) be compact connected Kähler of dimension $n \geq 2$, and let $A \in H^{1,1}(X, \mathbb{R})$. Assume*

$$\frac{(n-2)\pi}{2} < \hat{\theta} < \frac{n\pi}{2}, \quad \int_X ([\omega] + \sqrt{-1}A)^n \in e^{\sqrt{-1}\hat{\theta}} \mathbb{R}_{>0}.$$

A smooth solution $\alpha \in A$ of $\Theta_\omega(\alpha) = \hat{\theta}$ exists if and only if there is a smooth closed $\underline{\alpha} \in A$, called a $\mathcal{C}$ -subsolution, satisfying

$$\sum_{k \neq j} \arctan \lambda_k(\omega^{-1}\underline{\alpha}) > \hat{\theta} - \frac{\pi}{2} \quad \text{on } X, \quad j = 1, \dots, n.$$

Theorem 1.3 (Chu–Lee–Takahashi [2]). *Let X be compact connected Kähler of dimension $n \geq 2$, $A \in H^{1,1}(X, \mathbb{R})$, and $B \in \mathcal{K}_X$. Assume*

$$\frac{(n-2)\pi}{2} < \hat{\theta} < \frac{n\pi}{2}, \quad \int_X (B + \sqrt{-1}A)^n \in e^{\sqrt{-1}\hat{\theta}} \mathbb{R}_{>0}.$$

Put $q = n\pi/2 - \hat{\theta} \in (0, \pi)$ and

$$P_p(C, B; q) = \operatorname{Re}(C + \sqrt{-1}B)^p - \cot q \operatorname{Im}(C + \sqrt{-1}B)^p.$$

The following are equivalent:

- (i) *Some Kähler form $\omega \in B$ admits a smooth closed $\alpha \in A$ with $\Theta_\omega(\alpha) = \hat{\theta}$.*
- (ii) *Every Kähler form $\omega \in B$ admits such a form $\alpha = \alpha_\omega$.*
- (iii) *For every irreducible analytic $Y \subseteq X$ of dimension $1 \leq p \leq n$ and every $t \geq 0$,*

$$\int_Y P_p(A + tB, B; q) \geq 0, \quad \text{strictly if } p < n.$$

If X is projective, these are also equivalent to

$$\int_Y P_p(A, B; q) > 0 \quad \text{for every irreducible } Y \subsetneq X, \quad 1 \leq p = \dim Y < n.$$

2. THE SURFACE REDUCTION AND THE SUBSOLUTION CRITERION

Proof of Theorem 1.1. Put $c = \cot \hat{\theta}$. The phase equation implies

$$\alpha^2 + 2c\alpha \wedge \omega - \omega^2 = 0, \quad (\alpha + c\omega)^2 = (1 + c^2)\omega^2.$$

Since $\lambda_1 + \lambda_2 > 0$ at a solution,

$$\lambda_j + c = \frac{1 + \lambda_j^2}{\lambda_1 + \lambda_2} > 0.$$

Thus $\alpha + c\omega$ is Kähler. Conversely, a Kähler form $\beta_0 \in A + cB$ satisfies $\int_X \beta_0^2 = (1 + c^2) \int_X \omega^2$ by compatibility. Yau's theorem [8] gives $\beta = \beta_0 + \sqrt{-1}\partial\bar{\partial}u > 0$ with $\beta^2 = (1 + c^2)\omega^2$. For $\alpha = \beta - c\omega$,

$$\lambda_1 + \lambda_2 \geq 2\sqrt{1 + c^2} - 2c > 0.$$

The polynomial equation therefore has phase in $(0, \pi)$ and yields exactly $\hat{\theta}$. □

Lemma 2.1. *Fix $q \in (0, \pi)$. If a smooth real $(1, 1)$ form $\underline{\alpha}$ satisfies*

$$\sum_{k \neq j} \operatorname{arccot} \lambda_k(\omega^{-1}\underline{\alpha}) < q \quad \text{for every } j, \quad \operatorname{arccot} \lambda \in (0, \pi),$$

then, for $t \geq 0$ and $1 \leq p < n$, the form

$$\operatorname{Re}(\underline{\alpha} + t\omega + \sqrt{-1}\omega)^p - \cot q \operatorname{Im}(\underline{\alpha} + t\omega + \sqrt{-1}\omega)^p$$

is strictly positive on every complex p -plane.

Proof. In a unitary diagonal frame put $\phi_j = \operatorname{arccot}(\lambda_j + t)$. Each nonempty proper subset $I \subsetneq \{1, \dots, n\}$ has $0 < \sum_{j \in I} \phi_j < q$. Its wedge coefficient, up to a positive factor, is

$$\begin{aligned} & \operatorname{Re} \prod_{j \in I} (\lambda_j + t + \sqrt{-1}) - \cot q \operatorname{Im} \prod_{j \in I} (\lambda_j + t + \sqrt{-1}) \\ &= \frac{\prod_{j \in I} \sqrt{1 + (\lambda_j + t)^2}}{\sin q} \sin\left(q - \sum_{j \in I} \phi_j\right) > 0. \end{aligned}$$

Restriction to any p -plane is a nonzero sum of these coefficients with nonnegative squared-minor weights. $\square$

Proof of Theorem 1.2. A solution is a subsolution because $\arctan \lambda_j < \pi/2$. Conversely, set $q = n\pi/2 - \hat{\theta}$. The partial arctangent inequalities are precisely the hypotheses of Lemma 2.1. All proper-subvariety ray tests in Theorem 1.3 follow. For the top degree, compatibility and differentiation give

$$\int_X P_n(A, B; q) = 0, \quad \frac{d}{dt} \int_X P_n(A + tB, B; q) = n \int_X P_{n-1}(\underline{\alpha} + t\omega, \omega; q) \wedge \omega > 0.$$

The remaining ray test follows; Theorem 1.3 supplies the solution. $\square$

This proves sufficiency for the ordinary $\mathcal{C}$ -subsolution inequalities, without an additional pointwise phase hypothesis on $\underline{\alpha}$. The original Collins–Jacob–Yau existence theorem additionally assumed $\Theta_\omega(\underline{\alpha}) > (n-2)\pi/2$ [3, Theorem 1.2]; the argument above is instead a deduction from [2, Theorem 1.3].

Fix a smooth closed $\alpha_0 \in A$. The $\partial\bar{\partial}$ -lemma writes every representative as $\alpha_u = \alpha_0 + \sqrt{-1}\partial\bar{\partial}u$; $u \in C^\infty(X, \mathbb{R})$ is defined modulo constants.

Lemma 2.2. *For a smooth α_u , the linearization of Θ_ω is*

$$D\Theta_u(v) = \eta^{i\bar{j}} v_{i\bar{j}}, \quad \eta = \omega + \alpha_u \omega^{-1} \alpha_u > 0.$$

Solutions in the same class with the same real phase have potentials differing by a constant.

Proof. In a diagonal frame the coefficients are $(1 + \lambda_j^2)^{-1} > 0$. For solutions u, v , integration along $v + s(u - v)$ gives a linear uniformly elliptic equation for $u - v$ on compact X . The strong maximum principle applies. $\square$

The difficult implication in Theorem 1.3 is the imported construction of smooth solutions from numerical ray positivity [2, Sections 3–7]. Resolution and mass concentration produce local subsolution currents; smoothing and gluing preserve strict phase inequalities; the resulting data close the continuity argument. Strict positivity for each subvariety is not a uniform lower bound over all subvarieties. Producing the needed analytic margin is part of that construction.

For projective X , the removal of the ray parameter can be seen using a very ample class γ . With $P_0 = 1$,

$$\int_Y P_p(A + t\gamma, B; q) = \sum_{k=0}^p \binom{p}{k} t^{p-k} \int_Y P_k(A, B; q) \gamma^{p-k}.$$

General complete intersections identify the coefficients with endpoint tests on lower-dimensional cycles. They are positive, except for the zero top-degree constant term on X . This verifies a test family; the full test-family theorem gives existence [2, Section 2, Corollary 1.5]. The some/every-background equivalence gives separate solutions, not a common potential or estimates uniform over all $\omega \in B$.

3. CATEGORICAL STABILITY AND EQUALITY WALLS

Let X be a smooth projective surface over $\mathbb{C}$, with $b, B \in \text{NS}(X)_{\mathbb{R}}$ and B ample. Write $\text{Coh}(X)$ for coherent sheaves, $D^b \text{Coh}(X)$ for their bounded derived category, and $\mathcal{H}^j(E)$ for the cohomology sheaves of a complex E . For a nonzero torsion-free sheaf F , define

$$\mu_{b,B}(F) = \frac{(c_1(F) - \text{rk}(F)b) \cdot B}{\text{rk}(F)}.$$

The following full subcategories form a torsion pair in $\text{Coh}(X)$:

- $\mathcal{F}_{b,B}$ consists of torsion-free sheaves F such that $\mu_{b,B}(G) \leq 0$ for every nonzero subsheaf $G \subseteq F$.
- $\mathcal{T}_{b,B}$ consists of sheaves T such that $\mu_{b,B}(Q) > 0$ for every nonzero torsion-free quotient $T \twoheadrightarrow Q$.

Both contain 0; all torsion sheaves belong to $\mathcal{T}_{b,B}$. The tilted heart is the abelian category

$$\mathcal{A}_{b,B} = \left\{ E \in D^b \text{Coh}(X) : \begin{array}{l} \mathcal{H}^{-1}(E) \in \mathcal{F}_{b,B}, \quad \mathcal{H}^0(E) \in \mathcal{T}_{b,B}, \\ \mathcal{H}^j(E) = 0 \quad (j \neq -1, 0) \end{array} \right\}.$$

Here $[m]$ denotes the shift with $\mathcal{H}^j(E[m]) = \mathcal{H}^{j+m}(E)$. A sequence $0 \rightarrow K \rightarrow E \rightarrow Q \rightarrow 0$ is exact in $\mathcal{A}_{b,B}$ when it comes from a distinguished triangle $K \rightarrow E \rightarrow Q \rightarrow K[1]$ with $K, E, Q \in \mathcal{A}_{b,B}$; it need not be a sheaf exact sequence [1, Section 2].

Set $\text{ch}^b(E) = e^{-b} \text{ch}(E)$ and

$$\begin{aligned} Z_{b,B}(E) &= - \int_X e^{-(b+\sqrt{-1}B)} \text{ch}(E) \\ &= \frac{B^2}{2} \text{ch}_0^b(E) - \int_X \text{ch}_2^b(E) + \sqrt{-1}B \cdot \text{ch}_1^b(E). \end{aligned}$$

For $0 \neq E \in \mathcal{A}_{b,B}$, this charge lies in $\{z : \text{Im } z > 0\} \cup \mathbb{R}_{<0}$ [1, Corollary 2.1]. Take $\arg Z_{b,B}(E) \in (0, \pi]$ and define $\phi_{b,B}(E) = \pi^{-1} \arg Z_{b,B}(E) \in (0, 1]$. The pair $\sigma_{b,B} = (Z_{b,B}, \mathcal{A}_{b,B})$ is a Bridgeland stability condition [1, Section 2]; a nonzero $E \in \mathcal{A}_{b,B}$ is stable precisely when

$$\phi_{b,B}(K) < \phi_{b,B}(E) \quad \text{for every } 0 \neq K \subsetneq E \text{ in } \mathcal{A}_{b,B}.$$

Replacing $<$ by $\leq$ defines semistability. Shifts of stable objects remain stable; their real phases extend by $\phi(E[m]) = \phi(E) + m$. For a line bundle L , test stability in the heart on L when $(c_1(L) - b) \cdot B > 0$, and on $L[1]$ when $(c_1(L) - b) \cdot B \leq 0$ [7, Section 2.1].

Theorem 3.1 (Mäkelä, preprint [7]). *Let X be a smooth connected projective surface, L a line bundle, and $b, B \in \text{NS}(X)_{\mathbb{R}}$ with B ample. Set $A = c_1(L) - b$ and*

$$\Psi_C(A, B) = (C \cdot B)(B^2 - A^2) + 2(C \cdot A)(A \cdot B).$$

The following are equivalent:

- (i) $\Psi_C(A, B) \geq 0$ for every integral curve $C \subset X$;
- (ii) $L^{\otimes k}$ is $\sigma_{kb, kB}$ -stable for every integer $k \geq 1$;
- (iii) $L^{\otimes k}$ is $\sigma_{kb, kB}$ -stable for all sufficiently large integers k .

For the projective surface data of Theorem 3.1, suppose $s = A \cdot B > 0$. Then

$$c = \cot \hat{\theta} = \frac{B^2 - A^2}{2s}, \quad \Psi_C(A, B) = 2s(A + cB) \cdot C.$$

Together with $(A + cB)^2 = (1 + c^2)B^2 > 0$ and $(A + cB) \cdot B > 0$, the surface Kähler-cone criterion gives

$$\text{smooth dHYM solvability} \iff \Psi_C(A, B) > 0 \text{ for every integral curve } C.$$

Joint scaling preserves the equation exactly: $\Theta_{k\omega}(k\alpha) = \Theta_{\omega}(\alpha)$. It does not preserve finite-scale phase comparisons with arbitrary subobjects.

Proposition 3.2. *Let X, L, b, B, A be as in Theorem 3.1, with $s = A \cdot B > 0$. For a fixed nonzero effective divisor C , and all sufficiently large integers k , $L^k(-C) \subset L^k$ is a subobject in $\mathcal{A}_{kb, kB}$. Its strict stability inequality is*

$$\Psi_C(A, B) - \frac{C^2}{k}s > 0.$$

Proof. For large k , the slopes are $k^2s > 0$ and $k^2s - kB \cdot C > 0$. Thus both line bundles belong to $\mathcal{F}_{kb, kB}$, as does their torsion quotient $L^k|_C$. The sheaf exact sequence is therefore also exact in $\mathcal{A}_{kb, kB}$. Put $x = (B^2 - A^2)/2$. Their charges are

$$\begin{aligned} Z_{kb, kB}(L^k) &= k^2(x + \sqrt{-1}s), \\ Z_{kb, kB}(L^k(-C)) &= k^2(x + \sqrt{-1}s) + kA \cdot C - C^2/2 - \sqrt{-1}kB \cdot C. \end{aligned}$$

The two imaginary parts are positive. The condition $\text{Im}(Z(L^k(-C))\overline{Z(L^k)}) < 0$ is exactly the stated inequality after division by $k^3/2$. $\square$

Letting $k \rightarrow \infty$ yields only $\Psi_C \geq 0$. The converse in Theorem 3.1 uses Mäkelä's effective-divisor detection theorem [7, Theorem 1.1, Corollary 1.5]: non-stability is detected by an effective divisor with $C^2 < 0$. Such a divisor need not be irreducible. The correction above is then strictly positive when $s > 0$ and $\Psi_C \geq 0$; shifted objects give the analogous argument for $s < 0$. For $s = 0$, $L[1]$ is stable by [7, Corollary 7.2]. Theorem 3.1 is a preprint result, not a published higher-dimensional correspondence.

Example 3.3 ([5], Example 2.11). On $X = \text{Bl}_p \mathbb{P}^2$, let H be the pullback of a line and E the exceptional curve. Set

$$B = (2H - E)/\sqrt{3}, \quad b = 0, \quad L = \mathcal{O}_X(H).$$

Then $\hat{\theta} = \pi/2$ and $\Psi_C = 4(C \cdot H)/\sqrt{3} \geq 0$, with $\Psi_E = 0$. Thus every L^k is $\sigma_{0,kB}$ -stable, but no smooth dHYM solution exists: H is nef and big, not Kähler.

Problem 3.4. For smooth projective X of dimension $n \geq 3$, a line bundle $L, b, B \in \text{NS}(X)_{\mathbb{R}}$ with B ample, set $A = c_1(L) - b$. Assume a compatible supercritical phase $\hat{\theta}$ and put $q = n\pi/2 - \hat{\theta}$. Determine geometric hypotheses and a family of Bridgeland stability conditions on $D^b \text{Coh}(X)$ whose (possibly asymptotic) stability of L is equivalent to the following inequalities for every irreducible $Y \subsetneq X$:

$$\int_Y (\text{Re}(A + \sqrt{-1}B)^p - \cot q \text{Im}(A + \sqrt{-1}B)^p) > 0 \quad (1 \leq p = \dim Y < n).$$

Problem 3.4 refines the categorical dHYM programme, the holomorphic counterpart of Thomas–Yau stability [4, Conjectures 1.2–1.3]. In the projective supercritical range, the numerical existence question is already answered by Theorem 1.3. What remains is to realize its tests by actual destabilizing objects, with a specified stability theory and a treatment of equality walls. The surface example rules out identifying unqualified all-scale stability with smooth solvability. Away from all walls $\Psi_C = 0$, the surface weak and strict criteria agree [5, 7].

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